UID EXP – Student Performance Visualization

AIM:

The aim is to create data visualizations such as pie charts and bar graphs to analyze student performance using JavaScript.

PROCEDURE:

Step 1: Create HTML File (index.html)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Performance Visualization</title>

<style>
body {
   font-family: Arial;
   text-align: center;
   margin: 50px;
}
canvas {
   margin: 20px auto;
}
</style>
</head>

<body>

<h1>Student Performance System</h1>

<canvas id="pieChart" width="400" height="400"></canvas>
<canvas id="barChart" width="400" height="400"></canvas>

<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script src="script.js"></script>
```

```
</body>
</html>
```

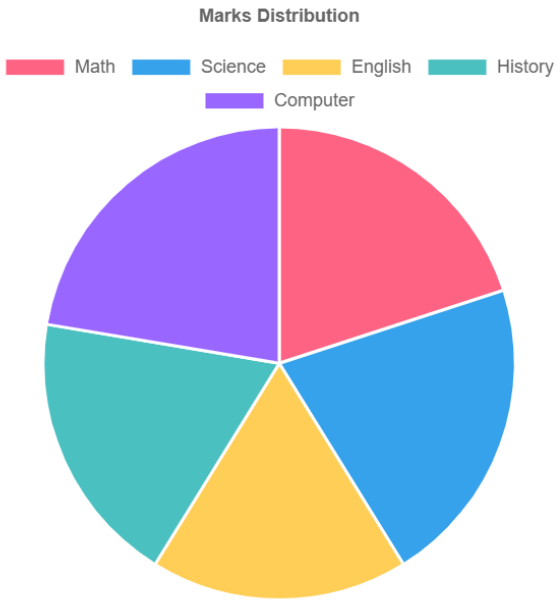
Step 2: Create JavaScript File (script.js)

```
// Student performance data
const studentData = {
labels: ['Math', 'Science', 'English', 'History', 'Computer'],
datasets: [{
  label: 'Marks',
  data: [85, 90, 75, 80, 95],
  backgroundColor: [
    '#FF6384',
    '#36A2EB',
    '#FFCE56',
    '#4BC0C0',
    '#9966FF'
  ]
}]
};

// PIE CHART
const ctxPie = document.getElementById('pieChart').getContext('2d');
new Chart(ctxPie, {
type: 'pie',
data: studentData,
options: {
  responsive: true,
  plugins: {
    title: {
      display: true,
      text: 'Marks Distribution'
    }
  }
}
});

// BAR CHART
const ctxBar = document.getElementById('barChart').getContext('2d');
new Chart(ctxBar, {
type: 'bar',
data: studentData,
options: {
  responsive: true,
```

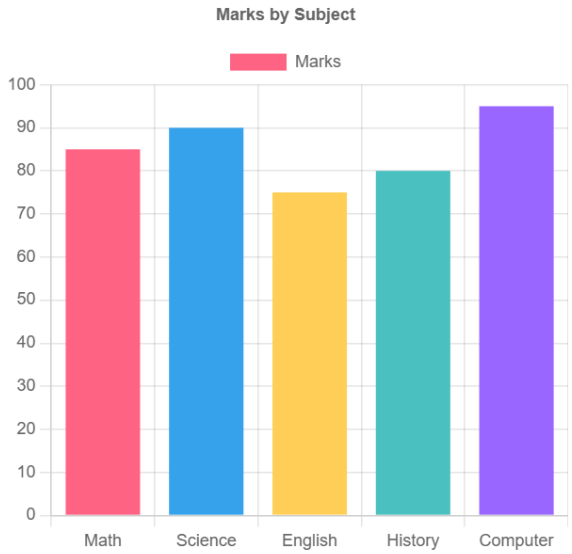
```
    plugins: {
        title: {
            display: true,
            text: 'Marks by Subject'
        }
    },
    scales: {
        y: {
            beginAtZero: true
        }
    }
}
});
```

OUTPUT:

Pie chart showing distribution of marks across subjects.



Bar chart comparing marks for each subject.



RESULT:

Thus, the student performance visualization system was successfully created using JavaScript and Chart.js.